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Otsubo

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(54) **CIRCUIT MODULE**

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(30) **Foreign Application Priority Data**

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H01R 12/52 (2011.01)
H05K 1/18 (2006.01)

(52) **U.S. Cl.**
CPC ***H01R 12/52*** (2013.01); ***H05K 1/181*** (2013.01); *H05K 2201/042* (2013.01); *H05K 2201/10234* (2013.01); *H05K 2201/10318* (2013.01); *H05K 2201/10522* (2013.01); *H05K 2201/10545* (2013.01)

(58) **Field of Classification Search**
CPC H01R 12/52
USPC 174/255
See application file for complete search history.

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(57) **ABSTRACT**

A circuit module includes an upper circuit board, a lower circuit board, a first conductor member, a second conductor member, and a first component. The upper circuit board has a first upper main surface and a first lower main surface. The lower circuit board has a second upper main surface and a second lower main surface, is disposed below the upper circuit board, and overlaps the upper circuit board as viewed in the downward direction. The first component is mounted on the first lower main surface or the second upper main surface. The first conductor member is a metal pin and is connected to the first lower main surface. The second conductor member is a metal pin and is connected to the second upper main surface. The second conductor member is electrically connected to the first conductor member.

9 Claims, 9 Drawing Sheets

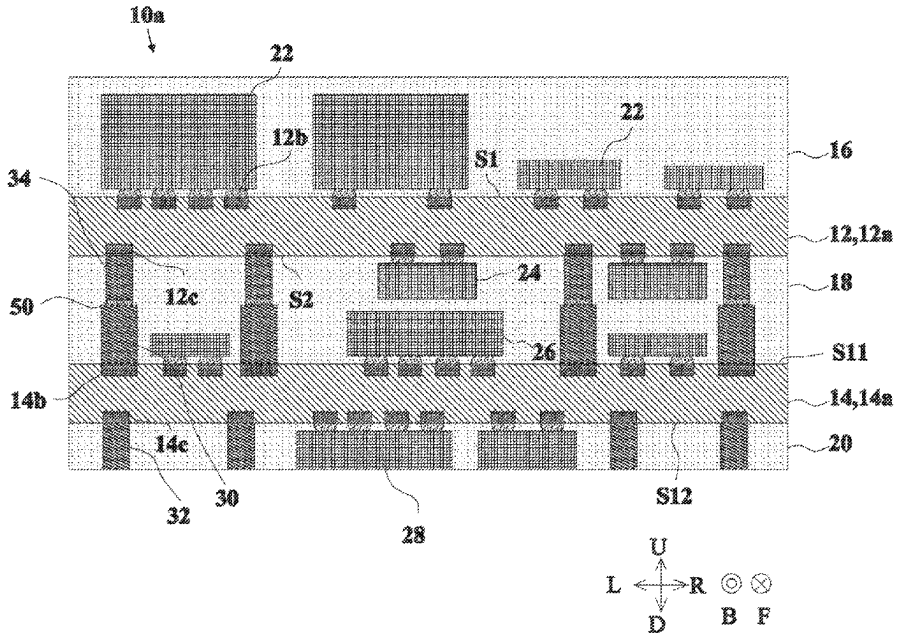

FIG. 1

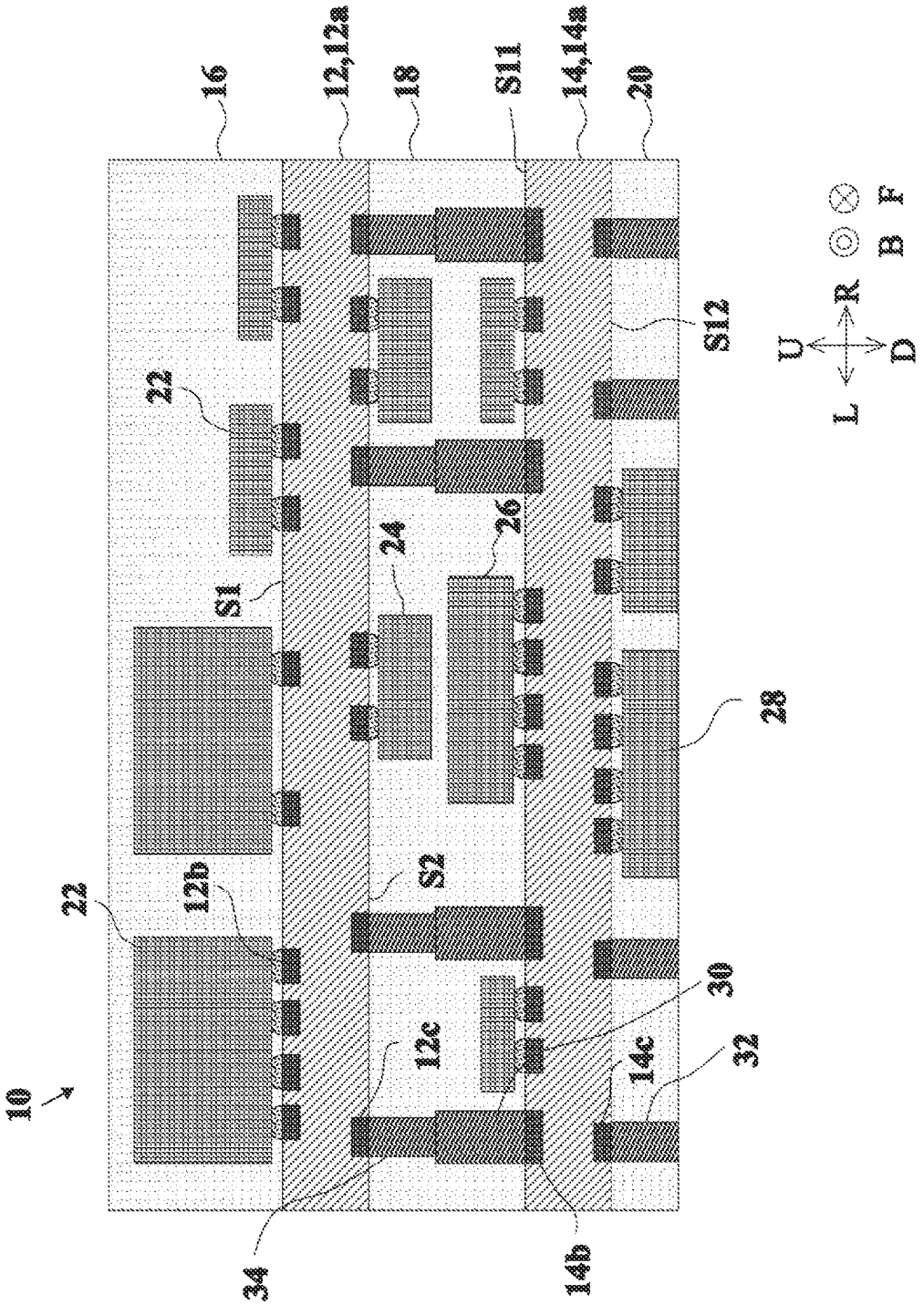
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FIG. 2

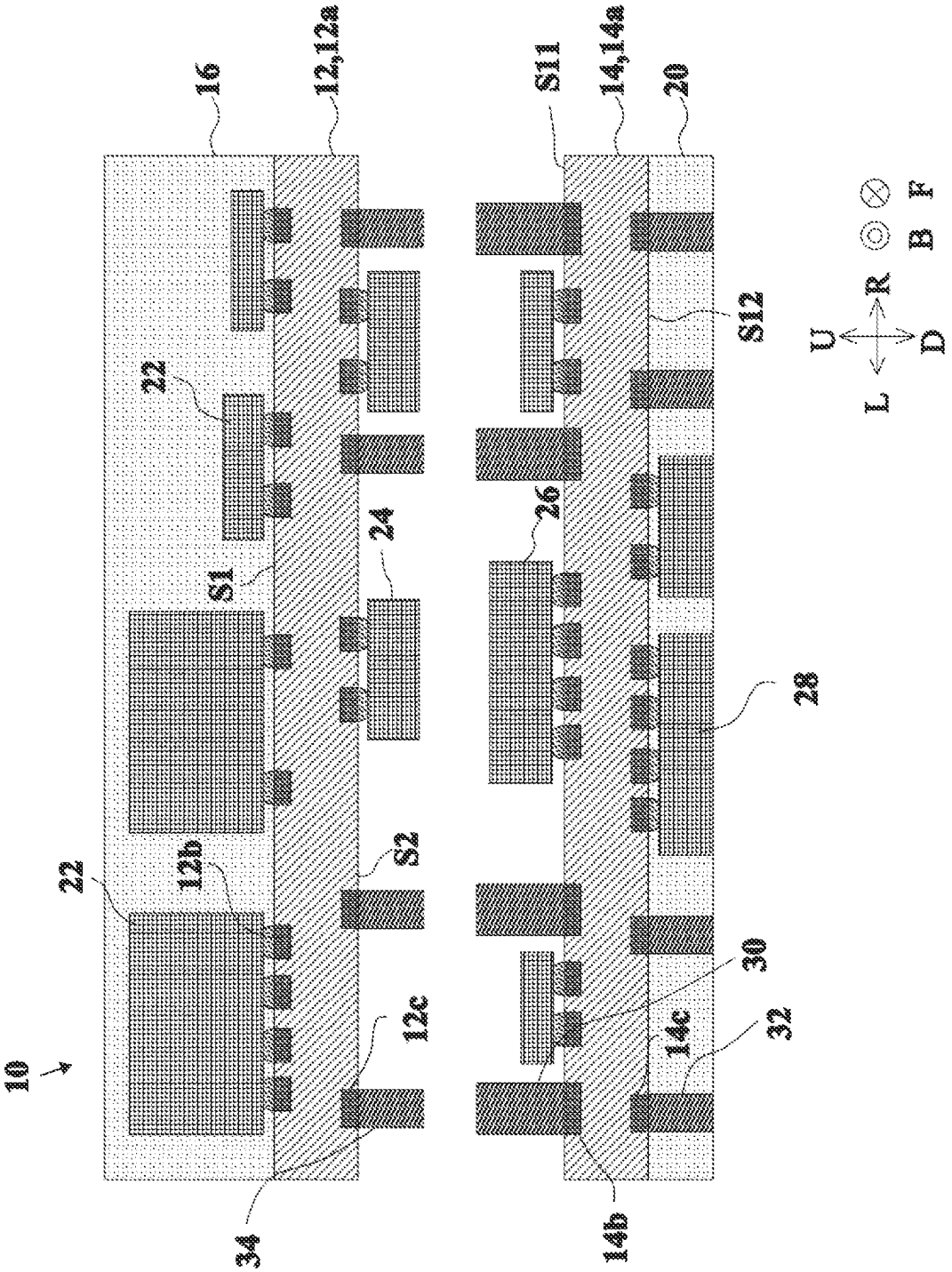
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FIG. 3

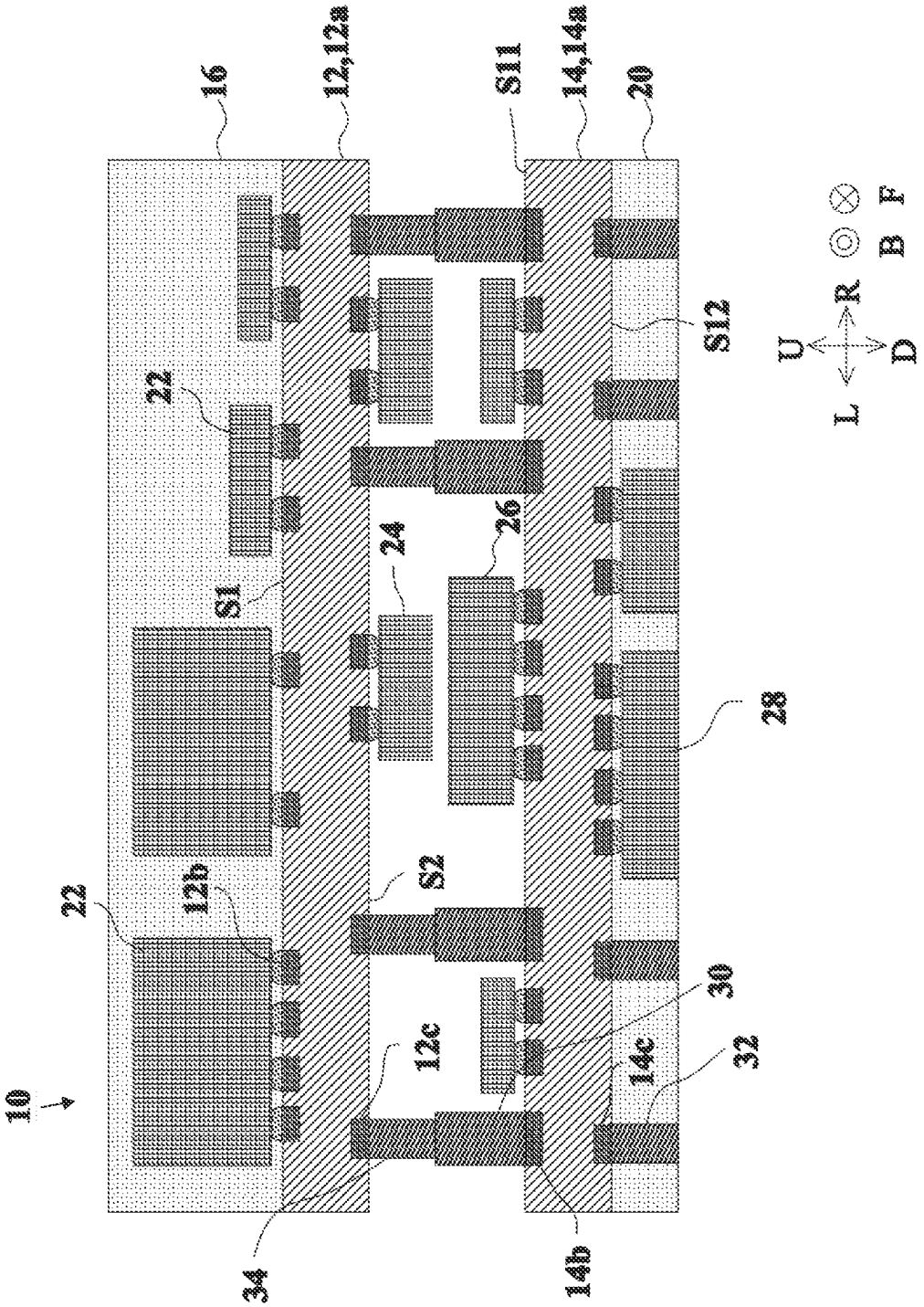
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FIG. 4

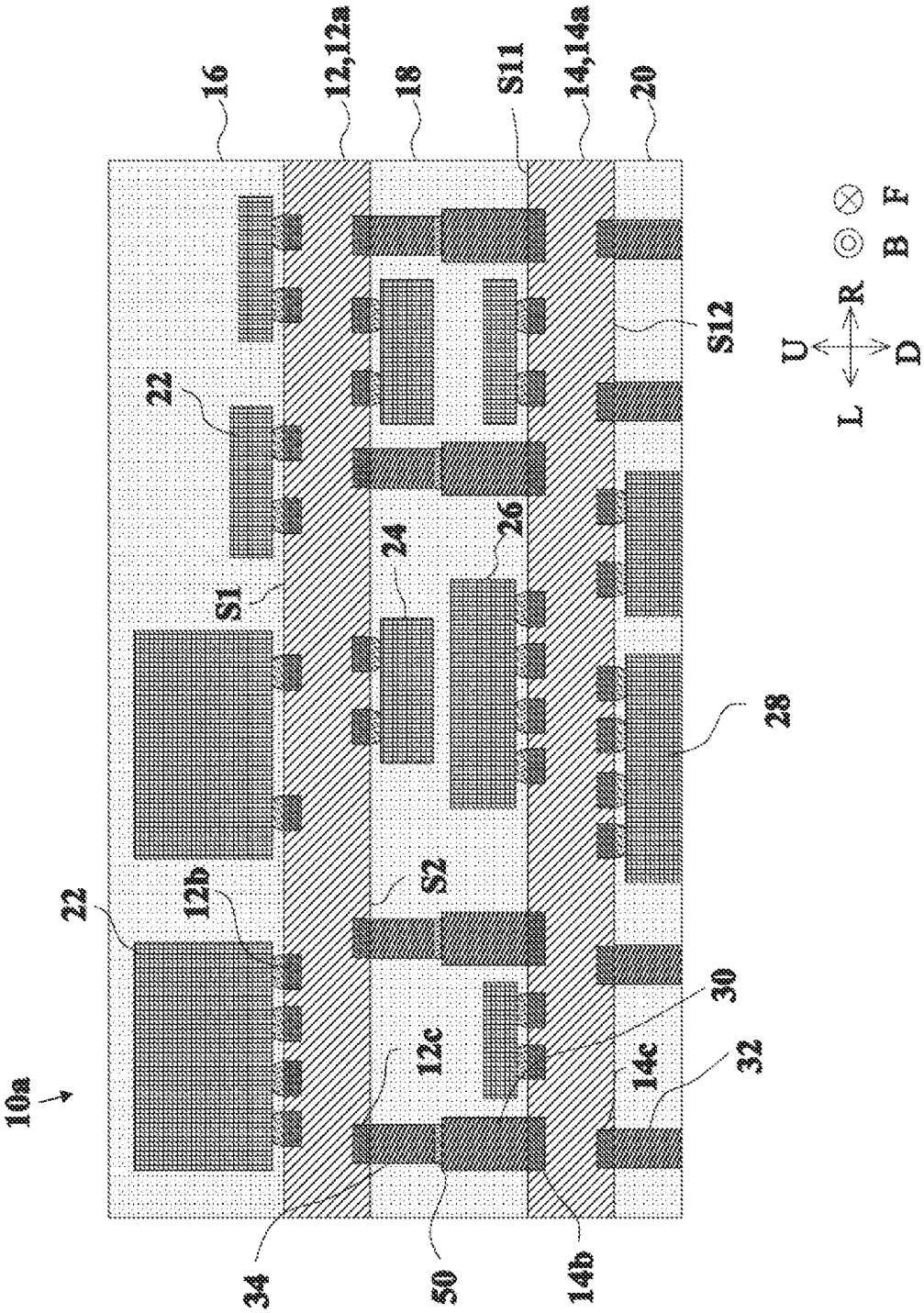
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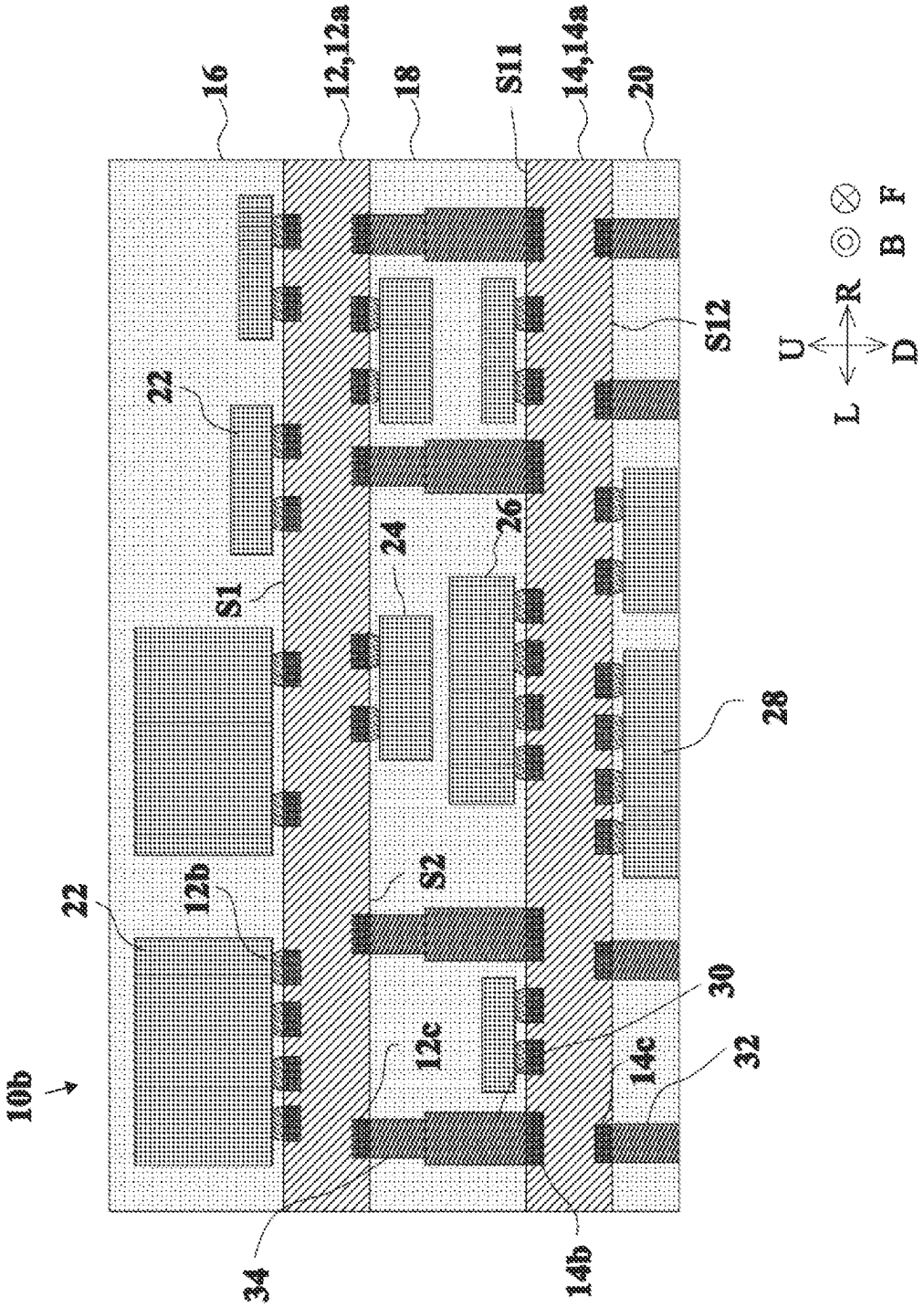
FIG. 5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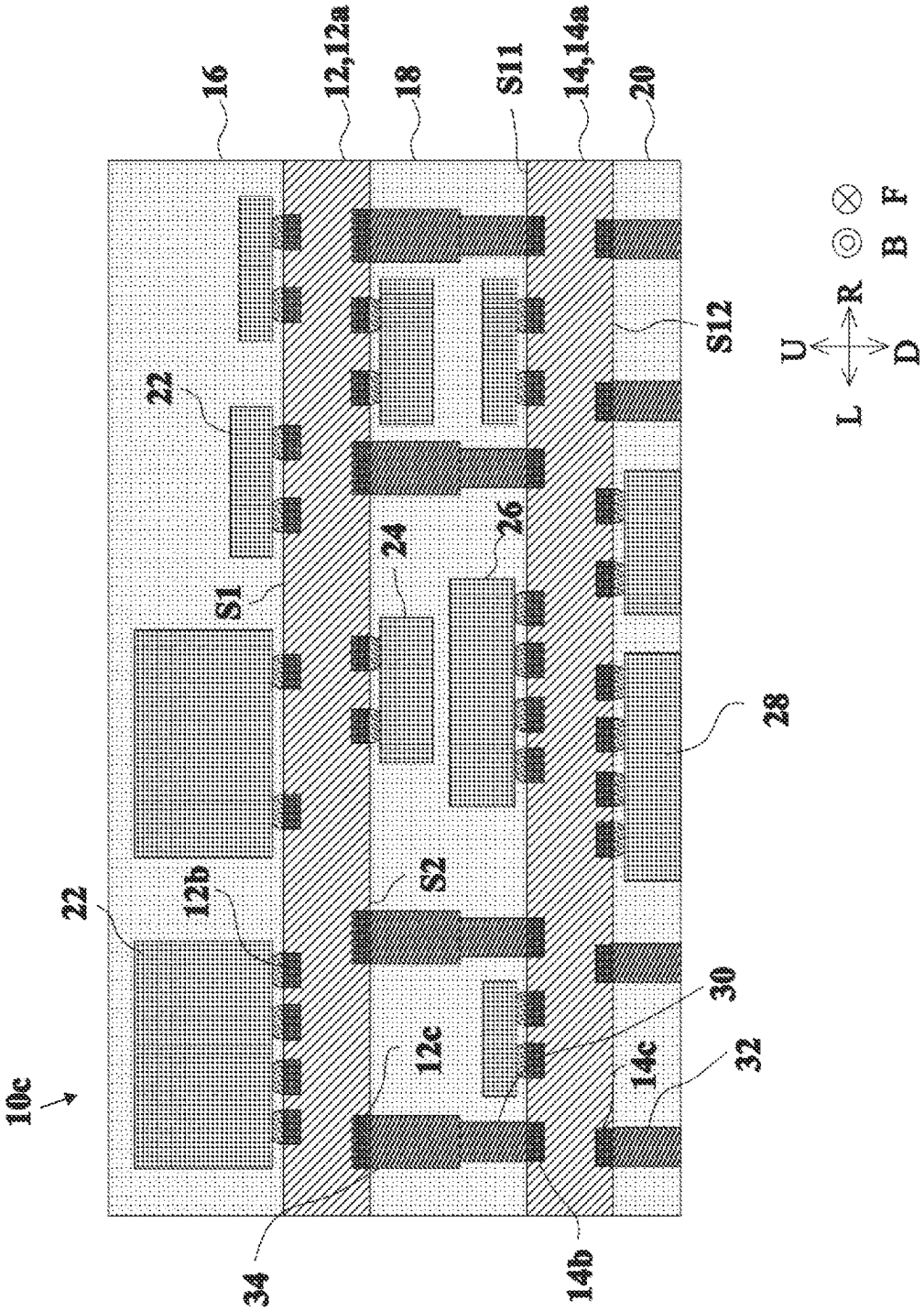
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FIG. 6

FIG. 7

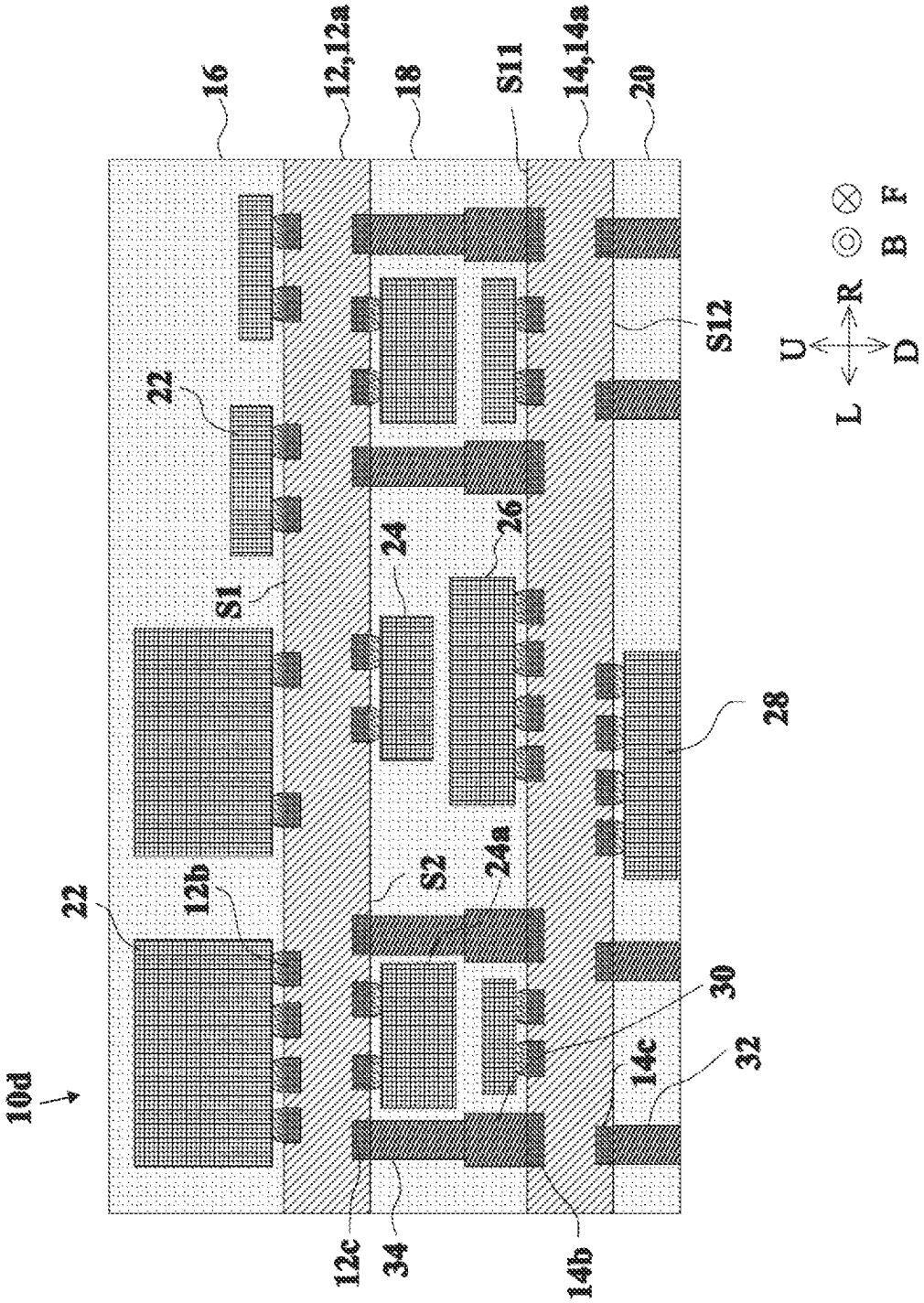
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FIG. 8

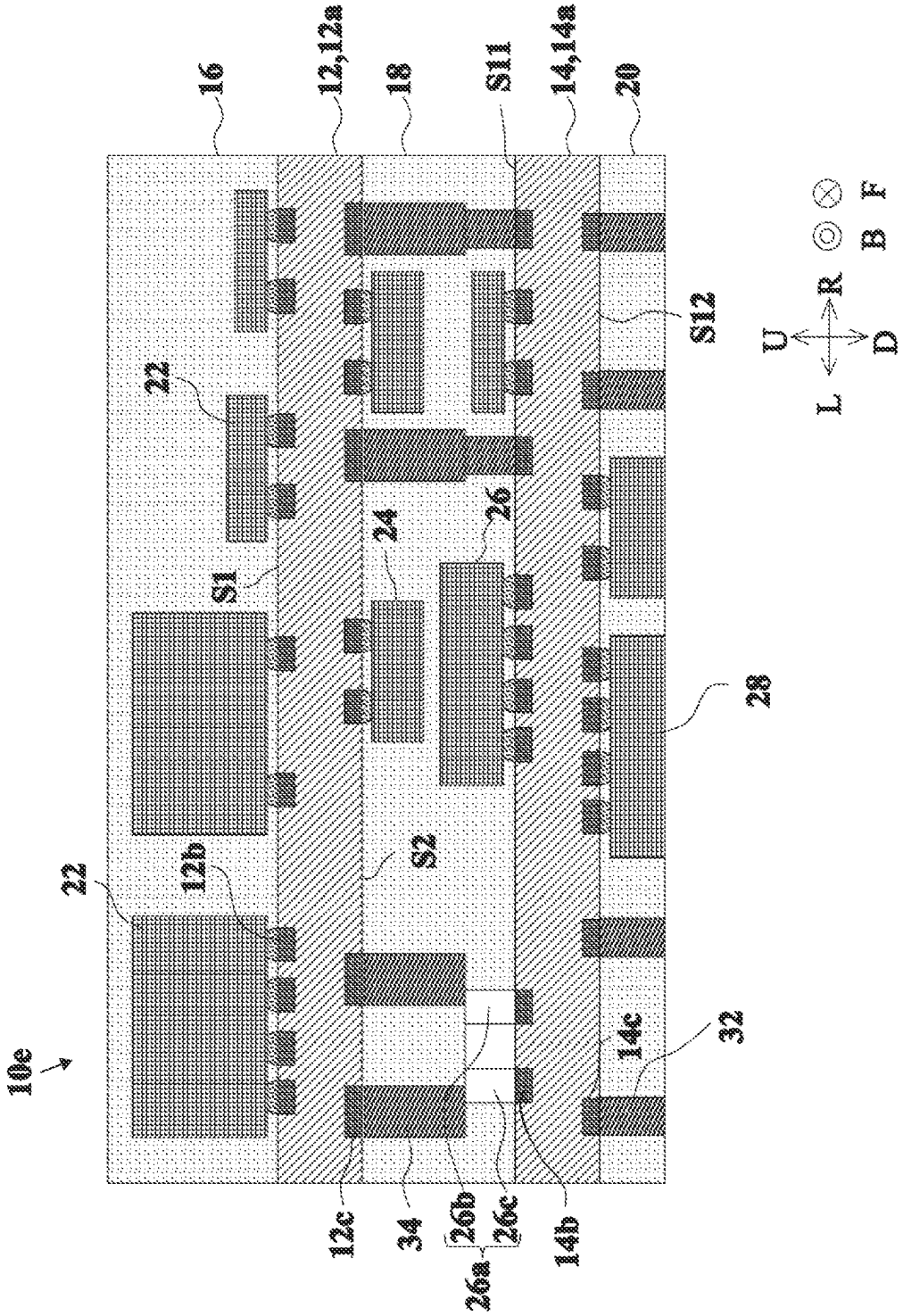
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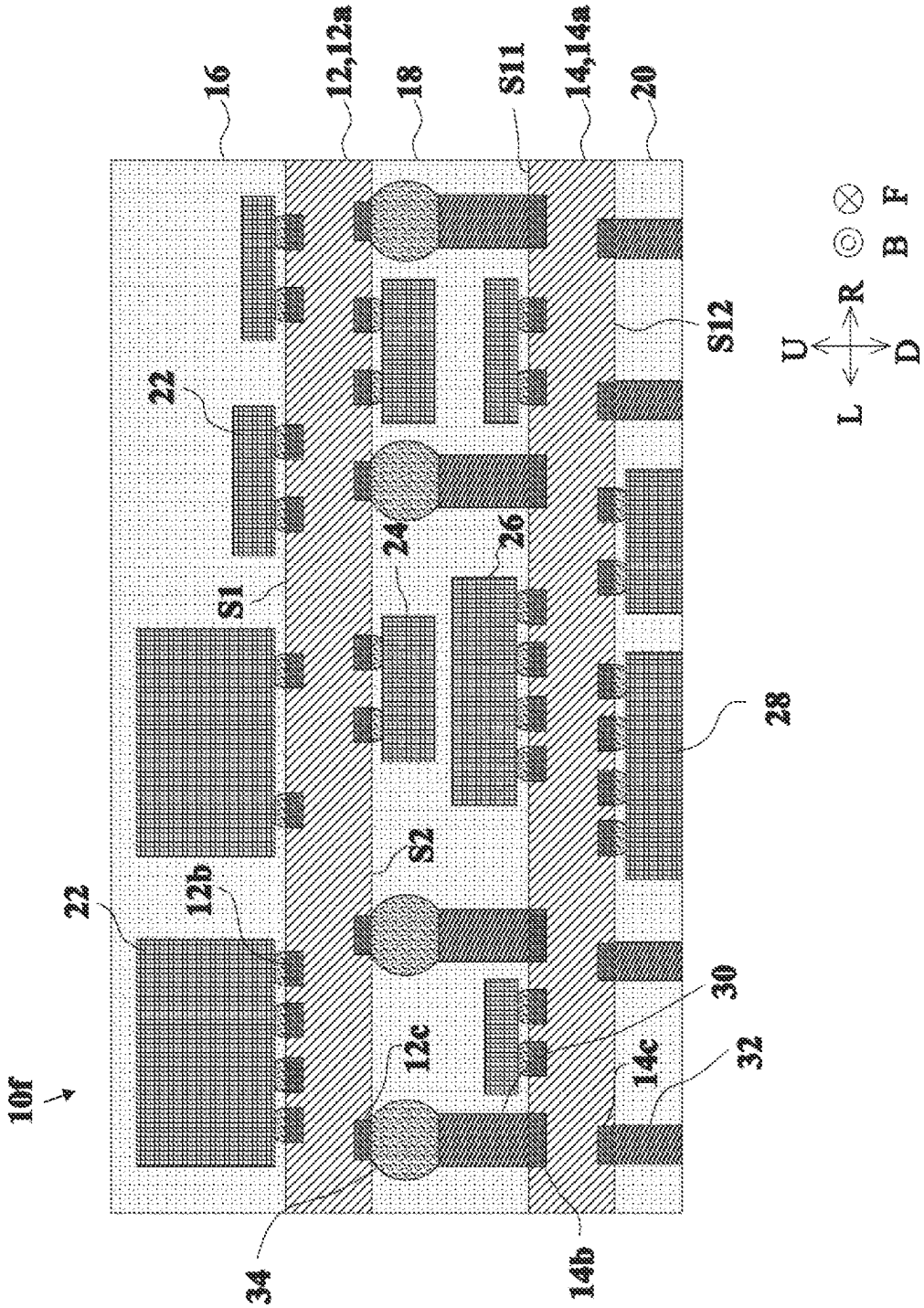
FIG. 9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CIRCUIT MODULE

CROSS REFERENCE TO RELATED APPLICATION

This application claims priority from Japanese Patent Application No. 2022-167401 filed on Oct. 19, 2022. The content of this application is incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

The present disclosure relates to an electronic device including an upper circuit board and a lower circuit board.

2. Description of the Related Art

As a disclosure related to a conventional circuit module, for example, a semiconductor device package described in U.S. Pat. No. 11,133,244 is known. This composite substrate includes a first substrate, a second substrate, and a conductive structure. The first substrate is disposed on the second substrate. The conductive structure is a pin extending along an up-down axis. The conductive structure is mounted on a lower main surface of the first substrate and an upper main surface of the second substrate. Due to this, the conductive structure electrically connects the first substrate and the second substrate.

BRIEF SUMMARY OF THE DISCLOSURE

Regarding the semiconductor device package described in U.S. Pat. No. 11,133,244, there is a demand for mounting more components between the first substrate and the second substrate. In this case, an interval between the first substrate and the second substrate increases. Since the conductive structure becomes long, the possibility that the conductive structure falls becomes high.

A possible benefit of the present disclosure is to provide a circuit module capable of mounting more components between an upper circuit board and a lower circuit board.

A circuit module according to an embodiment of the present disclosure includes: an upper circuit board, a lower circuit board, a first conductor member, a second conductor member, and a first component, in which the upper circuit board has a first upper main surface and a first lower main surface, the lower circuit board has a second upper main surface and a second lower main surface, is disposed below the upper circuit board, and overlaps the upper circuit board as viewed in a downward direction, the first component is mounted on the first lower main surface or the second upper main surface, the first conductor member is a metal pin or a solder bump, and is connected to the first lower main surface, the second conductor member is a metal pin and is connected to the second upper main surface, the second conductor member is electrically connected to the first conductor member, and a length of the second conductor member in an up-down direction is longer than a maximum value of a length of the second conductor member in an orthogonal direction orthogonal to the up-down direction.

According to the circuit module according to the present disclosure, more components can be mounted between the upper circuit board and the lower circuit board.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. **1** is a cross-sectional view of a circuit module **10**;

FIG. **2** is a cross-sectional view illustrating a manufacturing process of the circuit module **10**;

FIG. **3** is a cross-sectional view illustrating a manufacturing process of the circuit module **10**;

FIG. **4** is a cross-sectional view of a circuit module **10***a;*

FIG. **5** is a cross-sectional view of a circuit module **10***b;*

FIG. **6** is a cross-sectional view of a circuit module **10***c;*

FIG. **7** is a cross-sectional view of a circuit module **10***d;*

FIG. **8** is a cross-sectional view of a circuit module **10***e*; and

FIG. **9** is a cross-sectional view of a circuit module **10***f.*

DETAILED DESCRIPTION OF THE DISCLOSURE

Embodiments

[Structure of Circuit Module **10**]

The structure of the circuit module **10** according to an embodiment of the present disclosure will be described below with reference to the drawings. FIG. **1** is a cross-sectional view of a circuit module **10**. In FIG. **1**, reference signs are given only to a component **22**, a first component **24**, a second component **26**, a component **28**, a first conductor member **34**, a second conductor member **30**, a third conductor member **32**, a mounting electrode **12***b*, a mounting electrode **12***c*, a mounting electrode **14***b*, and a mounting electrode **14***c* that are representative among a plurality of the components **22**, a plurality of the first components **24**, a plurality of the second components **26**, a plurality of the components **28**, a plurality of the first conductor members **34**, a plurality of the second conductor members **30**, a plurality of the third conductor members **32**, a plurality of the mounting electrodes **12***b*, a plurality of the mounting electrodes **12***c*, a plurality of the mounting electrodes **14***b*, and a plurality of the mounting electrodes **14***c*.

Hereinafter, as illustrated in FIG. **1**, a direction from an upper circuit board **12** toward a lower circuit board **14** is defined as a downward direction of an up-down axis that coincides with an up-down direction. Axes orthogonal to the up-down axis are defined as a front-back axis and a left-right axis. The up-down axis, the front-back axis, and the left-right axis are orthogonal to one another. The front-back axis coincides with an axis extending along an axis perpendicular to the drawing sheet surface of FIG. **1**. The left-right axis coincides with the left-right axis of the drawing sheet surface of FIG. **1**. However, the up-down axis, the left-right axis, and the front-back axis in the present specification are directions defined for convenience of description, and need not coincide with the up-down axis, the left-right axis, and the front-back axis in use of the circuit module **10**. In each drawing, the upward direction and the downward direction may be interchanged, the left direction and the right direction may be interchanged, and the front direction and the back direction may be interchanged.

The circuit module **10** is used for a wireless communication terminal such as a smartphone. The circuit module **10** includes, for example, an integrated circuit (IC) such as a power amplifier (PA) or a low noise amplifier (LNA), and a coil, a capacitor, and the like. The coil, the capacitor, and the like are impedance matching circuit elements for each IC. The circuit module **10** has a cuboid shape. As illustrated in FIG. **1**, the circuit module **10** includes the upper circuit board **12**, the lower circuit board **14**, a sealing resin **16**, a sealing resin **18**, a sealing resin **20**, a plurality of components **22**, a plurality of first components **24**, a plurality of second components **26**, a plurality of components **28**, a plurality of

first conductor members **34**, a plurality of second conductor members **30**, and a plurality of third conductor members **32**.

As illustrated in FIG. **1**, the upper circuit board **12** has a first upper main surface S**1** and a first lower main surface S**2**. More specifically, the upper circuit board **12** includes a board body **12***a*, the plurality of mounting electrodes **12***b*, and the plurality of mounting electrodes **12***c*. The board body **12***a* has a plate shape having an upper main surface and a lower main surface. The board body **12***a* has a rectangular shape having two sides along the front-back axis and two sides extending along the left-right axis as viewed in the downward direction. The board body **12***a* has a multilayer structure. An electric circuit not illustrated is provided inside the board body **12***a*. The material of the board body **12***a* is, for example, a glass epoxy or a low temperature co-fired ceramics (LTCC) board, or the like.

As illustrated in FIG. **1**, the plurality of mounting electrodes **12***b* are provided on the upper main surface of the board body **12***a*. The plurality of mounting electrodes **12***c* are provided on the lower main surface of the board body **12***a*. The plurality of mounting electrodes **12***b* and the plurality of mounting electrodes **12***c* have a rectangular shape or a circular shape as viewed in the downward direction. The plurality of mounting electrodes **12***b* and the plurality of mounting electrodes **12***c* have a structure in which, for example, Ni plating and Au plating are applied to the surface of a copper-thick film electrode.

Here, the first upper main surface S**1** of the upper circuit board **12** includes the upper main surface of the board body **12***a* and the plurality of mounting electrodes **12***b*. The first lower main surface S**2** of the upper circuit board **12** includes the lower main surface of the board body **12***a* and the plurality of mounting electrodes **12***c*.

As illustrated in FIG. **1**, the lower circuit board **14** has a second upper main surface S**11** and a second lower main surface S**12**. More specifically, the lower circuit board **14** includes a board body **14***a*, the plurality of mounting electrodes **14***b*, and the plurality of mounting electrodes **14***c*. The board body **14***a* has a plate shape having an upper main surface and a lower main surface. The board body **14***a* has a rectangular shape having two sides extending along the front-back axis and two sides extending along the left-right axis as viewed in the downward direction. The board body **14***a* has a multilayer structure. An electric circuit not illustrated is provided inside the board body **14***a*. The material of the board body **14***a* is, for example, a glass epoxy or an LTCC board, or the like.

As illustrated in FIG. **1**, the plurality of mounting electrodes **14***b* are provided on the upper main surface of the board body **14***a*. The plurality of mounting electrodes **14***c* are provided on the lower main surface of the board body **14***a*. The plurality of mounting electrodes **14***b* and the plurality of mounting electrodes **14***c* have a rectangular shape or a circular shape as viewed in the up-down direction. The plurality of mounting electrodes **14***b* and the plurality of mounting electrodes **14***c* have a structure in which, for example, Ni plating and Au plating are applied to the surface of a copper thin film.

Here, the second upper main surface S**11** of the lower circuit board **14** includes the upper main surface of the board body **14***a* and the plurality of mounting electrodes **14***b*. The second lower main surface S**12** of the lower circuit board **14** includes the lower main surface of the board body **14***a* and the plurality of mounting electrodes **14***c*.

As illustrated in FIG. **1**, the lower circuit board **14** as described above is disposed below the upper circuit board **12**. The lower circuit board **14** overlaps the upper circuit board **12** as viewed in the downward direction. As viewed in the downward direction, the entire outer edge of the lower circuit board **14** overlaps the entire outer edge of the upper circuit board **12**.

The plurality of first conductor members **34** are metal pins. The plurality of first conductor members **34** have a rod shape extending along the up-down axis. The length in the up-down direction of the plurality of first conductor members **34** is longer than the maximum value of the length of the plurality of first conductor members **34** in the orthogonal direction orthogonal to the up-down direction. The plurality of first conductor members **34** are connected to the first lower main surface S**2**. Specifically, each of the upper ends of the plurality of first conductor members **34** is fixed to the plurality of mounting electrodes **12***c* by soldering. The material of the plurality of first conductor members **34** is, for example, copper.

The plurality of second conductor members **30** are metal pins. The plurality of second conductor members **30** have a rod shape extending along the up-down axis. Therefore, the length in the up-down direction of the plurality of second conductor members **30** is longer than the maximum value of the length of the plurality of second conductor members **30** in the orthogonal direction orthogonal to the up-down direction. Furthermore, the plurality of second conductor members **30** are thicker than the plurality of first conductor members **34**. That is, the maximum value of the length of the plurality of second conductor members **30** in the orthogonal direction orthogonal to the up-down direction is larger than the maximum value of the length of the plurality of first conductor members **34** in the orthogonal direction orthogonal to the up-down direction. In the present embodiment, the area of the upper ends of the plurality of second conductor members **30** is larger than the area of the lower ends of the plurality of first conductor members **34**. The plurality of second conductor members **30** are connected to the second upper main surface S**11**. Specifically, each of the lower ends of the plurality of second conductor members **30** is fixed to the plurality of mounting electrodes **14***b* by soldering.

Furthermore, the plurality of second conductor members **30** are disposed below the plurality of first conductor members **34**. The upper ends of the plurality of second conductor members **30** are in contact with the lower ends of the plurality of first conductor members **34**. Due to this, the plurality of second conductor members **30** are electrically connected to the plurality of first conductor members **34**. The material of the plurality of second conductor members **30** is, for example, copper. The plurality of first conductor members **34** and the plurality of second conductor members **30** having the above structure electrically connect the upper circuit board **12** and the lower circuit board **14**.

The plurality of third conductor members **32** are metal pins. The plurality of third conductor members **32** have a rod shape extending along the up-down axis. The length in the up-down direction of the plurality of third conductor members **32** is longer than the maximum value of the length of the plurality of third conductor members **32** in the orthogonal direction orthogonal to the up-down direction. The plurality of third conductor members **32** are connected to the second lower main surface S**12**. Specifically, each of the upper ends of the plurality of third conductor members **32** is fixed to the mounting electrode **14***c* of the lower circuit board **14** by soldering. The plurality of third conductor members **32** come into contact with a mounting electrode of a motherboard when the circuit module **10** is mounted on the motherboard. The material of the plurality of third conductor members **32** is, for example, copper.

As illustrated in FIG. **1**, the plurality of components **22** are mounted on the first upper main surface S**1** of the upper circuit board **12**. More specifically, the plurality of components **22** are mounted on the plurality of mounting electrodes **12***b*.

As illustrated in FIG. **1**, the plurality of first components **24** are mounted on the first lower main surface S**2** of the upper circuit board **12**. More specifically, the plurality of first components **24** are mounted on the plurality of mounting electrodes **12***c*.

As illustrated in FIG. **1**, the plurality of second components **26** are mounted on the second upper main surface S**11** of the lower circuit board **14**. More specifically, the plurality of second components **26** are mounted on the plurality of mounting electrodes **14***b*.

As illustrated in FIG. **1**, the plurality of components **28** are mounted on the second lower main surface S**12** of the lower circuit board **14**. More specifically, the plurality of components **28** are mounted on the plurality of mounting electrodes **14***c*. The plurality of components **22**, the plurality of first components **24**, the plurality of second components **26**, and the plurality of components **28** as described above are electronic components such as chip coils, chip capacitors, and semiconductor integrated circuits.

As illustrated in FIG. **1**, the sealing resin **18** covers the first lower main surface S**2** of the upper circuit board **12** and the second upper main surface S**11** of the lower circuit board **14**. The sealing resin **18** is provided between the upper circuit board **12** and the lower circuit board **14** so as to cover the surfaces of the plurality of first conductor members **34**, the plurality of second conductor members **30**, the plurality of first components **24**, and the plurality of second components **26**. The sealing resin **18** has a cuboid shape. As viewed in the downward direction, the entire outer edge of the sealing resin **18** overlaps the entire outer edge of the upper circuit board **12** and the entire outer edge of the lower circuit board **14**.

As illustrated in FIG. **1**, the sealing resin **16** covers the first upper main surface S**1** of the upper circuit board **12**. The sealing resin **16** covers the surfaces of the plurality of components **22**. The sealing resin **16** has a cuboid shape. As viewed in the up-down direction, the entire outer edge of the sealing resin **16** overlaps the entire outer edge of the upper circuit board **12** and the entire outer edge of the lower circuit board **14**.

As illustrated in FIG. **1**, the sealing resin **20** covers the second lower main surface S**12** of the lower circuit board **14**. The sealing resin **20** covers the surfaces of the plurality of third conductor members **32** and the plurality of components **28**. However, the lower ends of the plurality of third conductor members **32** and the lower surfaces of the plurality of components **28** are exposed from the sealing resin **20**. The sealing resin **20** has a cuboid shape. As viewed in the up-down direction, the entire outer edge of the sealing resin **20** overlaps the entire outer edge of the upper circuit board **12** and the entire outer edge of the lower circuit board **14**. The materials of the sealing resin **18**, the sealing resin **16**, and the sealing resin **20** as described above are, for example, epoxy resins.

The circuit module **10** as described above is produced by the following procedure. FIGS. **2** and **3** are cross-sectional views illustrating the manufacturing process of the circuit module **10**.

First, as illustrated in FIG. **2**, the plurality of first conductor members **34** are mounted on the plurality of mounting electrodes **12***c*. The plurality of second conductor members **30** are mounted on the plurality of mounting electrodes **14***b*.

Next, as illustrated in FIG. **3**, the lower ends of the plurality of first conductor members **34** are brought into contact with the upper ends of the plurality of second conductor members **30**. Then, as illustrated in FIG. **1**, the sealing resin **18** is injected between the upper circuit board **12** and the lower circuit board **14**.

Effects

According to the circuit module **10**, more components can be mounted between the upper circuit board **12** and the lower circuit board **14**. More specifically, regarding the semiconductor device package described in U.S. Pat. No. 11,133,244, there is a demand for mounting more components between the first substrate and the second substrate. In this case, an interval between the first substrate and the second substrate increases. In this case, since the conductive structure becomes long, the possibility that the conductive structure falls becomes high.

Therefore, in the circuit module **10**, the plurality of first conductor members **34** are connected to the first lower main surface S**2**. The plurality of second conductor members **30** are connected to the second upper main surface S**11**. Each of the plurality of second conductor members **30** is electrically connected to the plurality of first conductor members **34**. Due to this, the plurality of first conductor members **34** and the plurality of second conductor members **30** electrically connect the upper circuit board **12** and the lower circuit board **14**. Thus, the member electrically connecting the upper circuit board **12** and the lower circuit board **14** is divided into the plurality of first conductor members **34** and the plurality of second conductor members **30**, whereby the length of the plurality of first conductor members **34** and the length of the plurality of second conductor members **30** are shortened. As a result, the plurality of first conductor members **34** and the plurality of second conductor members **30** become less likely to fall. Therefore, in the circuit module **10**, since the interval between the upper circuit board **12** and the lower circuit board **14** can be increased, more components can be mounted between the upper circuit board **12** and the lower circuit board **14**.

According to the circuit module **10**, more components can be mounted between the upper circuit board **12** and the lower circuit board **14** also for the following reason. More specifically, the length of the plurality of second conductor members **30** in the up-down direction is longer than the maximum value of the length of the plurality of second conductor members **30** in the orthogonal direction orthogonal to the up-down direction. Due to this, the ratio of the plurality of second conductor members **30** to the second upper main surface S**11** decreases. As a result, many components can be mounted on the second upper main surface S**11**. As described above, according to the circuit module **10**, more components can be mounted between the upper circuit board **12** and the lower circuit board **14**.

According to the circuit module **10**, more components can be mounted between the upper circuit board **12** and the lower circuit board **14** also for the following reason. More specifically, the length of the plurality of first conductor members **34** in the up-down direction is longer than the maximum value of the length of the plurality of first conductor members **34** in the orthogonal direction orthogonal to the up-down direction. Due to this, the ratio of the plurality of first conductor members **34** to the first lower main surface

S2 decreases. As a result, many components can be mounted on the first lower main surface S2. As described above, according to the circuit module **10**, more components can be mounted between the upper circuit board **12** and the lower circuit board **14**.

In the circuit module **10**, the plurality of second conductor members **30** are thicker than the plurality of first conductor members **34**. That is, the maximum value of the length of the plurality of second conductor members **30** in the orthogonal direction orthogonal to the up-down direction is larger than the maximum value of the length of the plurality of first conductor members **34** in the orthogonal direction orthogonal to the up-down direction. Due to this, even if the plurality of first conductor members **34** are shifted in the front direction, the back direction, the left direction, or the right direction with respect to the plurality of second conductor members **30**, the plurality of first conductor members **34** and the plurality of second conductor members **30** are easily connected.

First Modification

Hereinafter, the circuit module **10***a* according to a first modification will be described with reference to the drawings. FIG. **4** is a cross-sectional view of a circuit module **10***a*.

The circuit module **10***a* is different from the circuit module **10** in including a plurality of solders **50**. The plurality of solders **50** are disposed between the plurality of first conductor members **34** and the plurality of second conductor members **30**. Due to this, the plurality of solders **50** fix the plurality of first conductor members **34** and the plurality of second conductor members **30**, and electrically connect the plurality of first conductor members **34** and the plurality of second conductor members **30**. The other structures of the circuit module **10***a* are the same as those of the circuit module **10**, and thus description will be omitted. The circuit module **10***a* has the same functions and effects as those of the circuit module **10**.

Second Modification

Hereinafter, the circuit module **10***b* according to a second modification will be described with reference to the drawings. FIG. **5** is a cross-sectional view of a circuit module **10***b*.

The circuit module **10***b* is different from the circuit module **10** in that the plurality of first conductor members **34** are short and the plurality of second conductor members **30** are long. Thus, in the circuit module **10***b*, the length, in the up-down direction, of the plurality of first conductor members **34** that are thin is shorter than the length, in the up-down direction, of the plurality of second conductor members **30** that are thick. The plurality of thin first conductor members **34** easily fall. Therefore, since the height of the plurality of thin first conductor members **34** in the up-down direction is low, the plurality of first conductor members **34** are less likely to fall. The other structures of the circuit module **10***b* are the same as those of the circuit module **10**, and thus description will be omitted. The circuit module **10***b* has the same functions and effects as those of the circuit module **10**.

Third Modification

Hereinafter, the circuit module **10***c* according to a third modification will be described with reference to the drawings. FIG. **6** is a cross-sectional view of a circuit module **10***c*.

The circuit module **10***c* is different from the circuit module **10** in that the plurality of first conductor members **34** are thicker than the plurality of second conductor members **30**. That is, the maximum value of the length of the plurality of first conductor members **34** in the orthogonal direction orthogonal to the up-down direction is larger than the maximum value of the length of the plurality of second conductor members **30** in the orthogonal direction orthogonal to the up-down direction. The other structures of the circuit module **10***c* are the same as those of the circuit module **10**, and thus description will be omitted. The circuit module **10***c* has the same functions and effects as those of the circuit module **10**.

Fourth Modification

Hereinafter, the circuit module **10***d* according to a fourth modification will be described with reference to the drawings. FIG. **7** is a cross-sectional view of a circuit module **10***d*.

The circuit module **10***d* is different from the circuit module **10** in that the plurality of first conductor members **34** are long and the plurality of second conductor members **30** are short. More specifically, in the circuit module **10***d*, the length of the plurality of thin first conductor members **34** in the up-down direction is longer than the length of the plurality of thick second conductor members **30** in the up-down direction. The height of the first component **24***a* in the up-down direction is higher than the height of the first component **24** in the up-down direction. The first conductor member **34** is disposed on each of the left and right of the first component **24***a*. The length of the plurality of first conductor members **34** in the up-down direction is longer than the height of the first component **24***a* in the up-down direction. Due to this, two first conductor members **34** can be brought close to the first component **24***a*. Therefore, the shortest distance between the second conductor member **30** and the second component **26** is shorter than the shortest distance between the first conductor member **34** and the first component **24***a*. Due to this, components can be mounted at high density. The other structures of the circuit module **10***d* are the same as those of the circuit module **10**, and thus description will be omitted. The circuit module **10***d* has the same functions and effects as those of the circuit module **10**.

Fifth Modification

Hereinafter, the circuit module **10***e* according to a fifth modification will be described with reference to the drawings. FIG. **8** is a cross-sectional view of a circuit module **10***e*.

The circuit module **10***e* is different from the circuit module **10** in further including a second component **26***a*. The second component **26***a* includes external electrodes **26***b* and **26***c*. The external electrodes **26***b* and **26***c* are mounted on the two mounting electrodes **14***b*. The two first conductor members **34** are connected to the external electrodes **26***b* and **26***c*. The other structures of the circuit module **10***e* are the same as those of the circuit module **10**, and thus description will be omitted. The circuit module **10***e* has the same functions and effects as those of the circuit module **10**.

Sixth Modification

Hereinafter, the circuit module **10***f* according to a sixth modification will be described with reference to the drawings. FIG. **9** is a cross-sectional view of a circuit module **10***f*.

The circuit module **10***f* is different from the circuit module **10** in that the first conductor member **34** is a solder bump. The solder bump has a spherical shape. The spherical shape also includes an ellipsoid. Therefore, in the solder bump, the position where the length has the maximum value in the orthogonal direction is below the upper end of the solder bump and above the lower end of the solder bump. In the present embodiment, the position where the length has the maximum value in the orthogonal direction is the center of the solder bump in the direction along the up-down axis. The other structures of the circuit module **10***f* are the same as those of the circuit module **10**, and thus description will be omitted. The circuit module **10***f* has the same functions and effects as those of the circuit module **10**.

OTHER EMBODIMENTS

The circuit module according to the present disclosure is not limited to the circuit modules **10** and **10***a* to **10***f*, and can be changed within the scope of the gist thereof. The structures of the circuit modules **10** and **10***a* to **10***f* may be arbitrarily combined.

The circuit modules **10** and **10***a* to **10***f* include the plurality of components **22**. However, the circuit modules **10** and **10***a* to **10***f* only need to include one or more components **22**. The circuit modules **10** and **10***a* to **10***f* include the plurality of first components **24**. However, the circuit modules **10** and **10***a* to **10***f* only need to include one or more first components **24**.

The circuit modules **10** and **10***a* to **10***f* include the plurality of second components **26**. However, the circuit modules **10** and **10***a* to **10***f* only need to include one or more second components **26**.

The circuit modules **10** and **10***a* to **10***f* include the plurality of components **28**. However, the circuit modules **10** and **10***a* to **10***f* only need to include one or more components **28**.

The circuit modules **10** and **10***a* to **10***f* include the plurality of first conductor members **34**. However, the circuit modules **10** and **10***a* to **10***f* only need to include one or more first conductor members **34**.

The circuit modules **10** and **10***a* to **10***f* include the plurality of second conductor members **30**. However, the circuit modules **10** and **10***a* to **10***f* only need to include one or more second conductor members **30**.

The circuit modules **10** and **10***a* to **10***f* include the plurality of third conductor members **32**. However, the circuit modules **10** and **10***a* to **10***f* only need to include one or more third conductor members **32**.

A shield member covering the front surface, the back surface, the left surface, the right surface, and the upper surface of the circuit module **10***f* may be provided.

The second component **26** is not an essential component. When the second component **26** is not provided, the first component **24** may be mounted on the second upper main surface S**11**.

In the circuit modules **10** and **10***a* to **10***f*, the plurality of third conductor members **32** may be mounted on the first upper main surface S**1**. In this case, the circuit board is arranged on the circuit modules **10** and **10***a* to **10***f*. The plurality of third conductor members **32** are mounted on the plurality of external electrodes of the circuit board.

The sealing resins **16**, **18**, and **20** are not essential components.

The present disclosure includes the following structure.

(1) A circuit module includes: an upper circuit board; a lower circuit board; a first conductor member; a second conductor member; and a first component, in which the upper circuit board has a first upper main surface and a first lower main surface, the lower circuit board has a second upper main surface and a second lower main surface, is disposed below the upper circuit board, and overlaps the upper circuit board as viewed in a downward direction, the first component is mounted on the first lower main surface or the second upper main surface, the first conductor member is a metal pin or a solder bump, and is connected to the first lower main surface, the second conductor member is a metal pin and is connected to the second upper main surface, the second conductor member is electrically connected to the first conductor member, and a length of the second conductor member in the up-down direction is longer than a maximum value of a length of the second conductor member in an orthogonal direction orthogonal to the up-down direction.

(2) The circuit module according to (1) further including a second component, in which the first component is mounted on the first lower main surface, and the second component is mounted on the second upper main surface.

(3) The circuit module according to (1) or (2) further including a sealing resin, in which the sealing resin covers the first lower main surface and the second upper main surface.

(4) The circuit module according to any of (1) to (3), in which the first conductor member is the solder bump, and a position where a length of the solder bump in the orthogonal direction has a maximum value is below an upper end of the solder bump and above a lower end of the solder bump.

(5) The circuit module according to any of (1) to (4), in which the first conductor member is a metal pin, and a length of the first conductor member in the up-down direction is longer than a maximum value of a length of the first conductor member in the orthogonal direction orthogonal to the up-down direction.

(6) The circuit module according to any of (1) to (5), in which a length of the first conductor member in the up-down direction is shorter than a length of the second conductor member in the up-down direction.

(7) The circuit module according to (2), in which a length of the second conductor member in the up-down direction is shorter than a length of the first conductor member in the up-down direction, and a shortest distance between the second conductor member and the second component is shorter than a shortest distance between the first conductor member and the first component.

What is claimed is:

1. A circuit module comprising:

an upper circuit board;

a lower circuit board;

a first conductor member;

a second conductor member;

a first component, and

a sealing resin,

wherein

the upper circuit board has a first upper main surface and a first lower main surface,

the lower circuit board has a second upper main surface and a second lower main surface, is disposed below the upper circuit board, and overlaps the upper circuit board as viewed in a downward direction of an up-down axis,

the first component is mounted on the first lower main surface or the second upper main surface,

the first conductor member is a metal pin or a solder bump, and is connected to the first lower main surface,

the second conductor member is a metal pin and is connected to the second upper main surface,

the second conductor member is electrically connected to the first conductor member,

a length of the second conductor member in an up-down direction is longer than a maximum value of a length of the second conductor member in an orthogonal direction orthogonal to the up-down direction, wherein the up-down direction extends along the up-down axis, and

the sealing resin covers the first lower main surface and the second upper main surface.

2. The circuit module according to claim **1**, further comprising a second component,

wherein

the first component is mounted on the first lower main surface, and

the second component is mounted on the second upper main surface.

3. The circuit module according to claim **2**, wherein

the length of the second conductor member in the up-down direction is shorter than a length of the first conductor member in the up-down direction, and

a shortest distance between the second conductor member and the second component is shorter than a shortest distance between the first conductor member and the first component.

4. The circuit module according to claim **2**, wherein

the first conductor member is the solder bump, and

a position where a length of the solder bump in the orthogonal direction has a maximum value is below an upper end of the solder bump and above a lower end of the solder bump.

5. The circuit module according to claim **2**, wherein

the first conductor member is a metal pin, and

a length of the first conductor member in the up-down direction is longer than a maximum value of a length of the first conductor member in the orthogonal direction.

6. The circuit module according to claim **2**, wherein a length of the first conductor member in the up-down direction is shorter than a length of the second conductor member in the up-down direction.

7. The circuit module according to claim **1**, wherein

the first conductor member is the solder bump, and

a position where a length of the solder bump in the orthogonal direction has a maximum value is below an upper end of the solder bump and above a lower end of the solder bump.

8. The circuit module according to claim **1**, wherein

the first conductor member is a metal pin, and

a length of the first conductor member in the up-down direction is longer than a maximum value of a length of the first conductor member in the orthogonal direction.

9. The circuit module according to claim **1**, wherein a length of the first conductor member in the up-down direction is shorter than a length of the second conductor member in the up-down direction.

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